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Beginner AI and the future of work

Why AI Won't Take Your Job: The Pyramid-to-Diamond Shift

Understand why AI-driven efficiency can increase demand for skilled human work, and which skills determine who benefits.

For knowledge workers, managers, and students trying to understand how AI is reshaping career ladders and what skills will matter next · 27 July 2026

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Overview

Most fears about AI and jobs assume a simple trade: AI does the work, a person loses the job. That fear comes from picturing an organization as a pyramid — a wide base of entry-level workers doing routine tasks, narrowing up through experienced staff, managers, and finally a small group of executives. This lesson argues that AI does not shrink that pyramid so much as reshape it into a diamond: fewer purely routine entry-level jobs, but a much larger band of people doing the judgment-heavy, AI-directing work that used to require years to reach.

The reshaping rests on two separate ideas. First, AI absorbs a large share of routine, well-defined execution work — first-draft writing, boilerplate code, applying a known procedure — which frees the humans who used to do that work to take on coordination, judgment, and goal-setting responsibilities earlier in their careers. Second, a 19th-century economic observation called Jevons Paradox suggests that making something more efficient to produce does not reduce total demand for it — it increases demand, because falling costs let people use more of it. Applied to labor, cheaper task execution via AI can mean organizations pursue more projects and more clients, which requires more people who can direct and supervise that expanded output, not fewer.

None of this is automatic. It only plays out for organizations that reinvest efficiency gains into growth rather than pure cost-cutting, and for individuals who actively build the skills the new roles require: flexibility, curiosity, creativity, and critical thinking. Staying static in a task AI now does better is the one strategy this lesson argues will not work.

KEY TAKEAWAYS

- ✓ Reframe the org chart from a pyramid (many entry-level roles, few executives) to a diamond (fewer purely routine entry roles, many more experience-level and senior roles).
- ✓ AI absorbing routine execution work does not eliminate the people who used to do it; it can move them into higher-judgment roles faster than a traditional career ladder would.
- ✓ Jevons Paradox: historically, making a resource or process more efficient increases total consumption of it rather than reducing it, because falling costs raise demand (steam engines and coal, 1865).
- ✓ Applied to AI and labor: as AI makes task execution cheaper, organizations that reinvest the savings into more projects and clients need more people to direct and supervise that work, not fewer.
- ✓ The effect only appears for growth-oriented organizations. A company that uses AI purely to cut headcount does not generate the extra demand that would pull people up into better roles.
- ✓ The skills that determine who gets the 'promotion': flexibility, curiosity, creativity, and critical thinking — all judgment skills, not execution skills.
- ✓ Moving up the diamond is not automatic for an individual either; it still requires real experience (e.g., internships) to build the judgment that senior-level roles depend on.

01 The traditional job pyramid

0:43

> Most organizations are shaped like a pyramid: a wide base of entry-level workers executing assigned tasks, narrowing through experienced staff, managers, and a small group of executives at the top.

Picture an organization's roles stacked as a pyramid. At the base are entry-level roles: people who are assigned a piece of work by someone above them and simply do it. This is often necessary but not always interesting work — it has to get done, and it falls to whoever is at this level.

Moving up, an experienced-level role owns a larger, more complete piece of work rather than just an assigned fragment of it. A software analogy: an entry-level programmer might write one module to a specification; an experienced one is responsible for an entire section of the system and how its pieces integrate with each other.

Above that sit senior staff and managers, whose defining responsibility is leading individuals — overseeing the people below them and making sure their work comes together into something coherent. At the top sit executives and the C-suite (the CEO and other chief officers), who lead the leaders: their view spans the whole organization rather than any single piece of work.

Each layer up has fewer people in it than the layer below, which is exactly why the traditional org chart is drawn as a pyramid: it is wide at the bottom and narrow at the top.

KEY POINTS

- → Entry-level roles execute a task assigned by someone else.
- → Experienced-level roles own a complete piece of work end-to-end, including how its parts fit together.
- → Senior and manager roles are defined by leading individuals, not by producing output directly.
- → Executives and the C-suite lead the leaders and think at the scale of the whole organization.
- → Headcount shrinks at each higher layer, which is what gives the pyramid its shape.

A software team mapped onto the pyramid

A junior developer (entry-level) implements a ticket someone else specified. A senior developer (experienced-level) owns an entire subsystem and decides how its modules connect. An engineering manager (senior/manager) leads the developers and is accountable for the team's delivery. A CTO (C-suite) sets technical direction across every team in the company.

02 From pyramid to diamond

2:51

> AI absorbs much of the routine entry-level work, shrinking the base of the pyramid, while the middle band of experience-level, AI-directing roles grows larger — turning the pyramid into a diamond.

If AI can perform a large share of what used to be entry-level, purely-execute work, that base of the pyramid gets thinner — but the people who used to occupy it do not simply disappear. A meaningful share of them move into what used to be considered experience-level work, because they can now direct a team of AI tools instead of needing years of practice to be trusted with a large chunk of work on their own.

The underlying shift is one of leverage. An experienced-level person who commands AI tools can accomplish what used to require an army of entry-level people executing tasks by hand. That person now owns a larger portion of the total work than an experienced person could before, because the tools multiply what one person can direct.

The same pattern repeats one level up: some experienced people move into roles that look more like today's senior positions, because directing an AI-augmented team is closer to a senior function (setting direction, checking quality) than to an individual-contributor one. The net shape is a diamond: a narrower entry-level base, and a wider middle and upper-middle band than the original pyramid had.

KEY POINTS

- → AI absorbing routine execution work narrows the entry-level base of the pyramid.
- → Former entry-level workers do not simply exit the workforce; many move into experience-level roles.
- → One person directing AI tools can accomplish what used to take a team of people doing manual execution.
- → The same upward shift can repeat a level higher, pushing experienced people toward senior-style responsibilities.
- → The resulting shape is a diamond: thinner at the very bottom, wider in the middle bands.

One person, an AI-augmented team

Where a marketing campaign used to require several people manually drafting copy, resizing assets, and scheduling posts, one experienced marketer directing AI tools for drafting and production can own the entire campaign end-to-end — the role that used to require a small team is now one person's expanded scope.

03 The promotion requires new skills, not just AI access

4:16

> Moving from entry-level to experience-level responsibility is not automatic just because AI removed the routine work; it requires judgment built from real experience, which is why internships and deliberate upskilling still matter.

Having AI do the grunt work does not, by itself, qualify someone to take on an experienced person's responsibilities. The skills that used to belong to more senior, decision-making functions — judging what's correct, what's risky, what's good enough to ship — now need to exist further down the org than before, and that takes deliberate development, not just tool access.

This is where experience still matters. You cannot skip straight from doing no independent work to owning a full piece of it; you need practice making judgment calls in a lower-stakes setting first. Internships and similar entry points remain valuable specifically because they are where someone can build that judgment before being trusted with a larger scope.

Organizations that want this shift to work need to invest in upskilling — deliberately improving what people know and can decide, not just handing them AI tools and expecting the judgment to appear on its own.

KEY POINTS

- → AI removing routine tasks does not automatically transfer the judgment that experience-level roles require.
- → Judgment is built through practice making decisions, typically in lower-stakes settings first.
- → Internships remain valuable as a structured way to gain that practice before taking on full ownership.
- → Organizations need to actively invest in upskilling, not assume AI access alone prepares people for bigger roles.

Report generation vs. report judgment

An intern using AI can generate a polished-looking market analysis in minutes. What they cannot yet do without experience is judge whether the AI picked the right comparison companies, whether a number that looks clean is actually wrong, or whether the conclusion matches what the client actually needs — that judgment is what the internship is meant to build.

04 Jevons Paradox: why efficiency can increase demand

5:44

> A observation from 1865: making a resource cheaper and more efficient to use does not reduce total demand for it, it increases demand, because falling costs make people use more of it.

In 1865, the economist William Stanley Jevons studied steam engines, which were becoming steadily more fuel-efficient. The intuitive prediction was that more efficient engines would need less coal, shrinking demand for coal and for coal miners. That is not what happened. As engines got more efficient, the cost of getting useful work out of a given amount of coal fell. Cheaper power meant steam engines became viable in more situations and affordable to more users, so total coal consumption rose rather than fell. This is Jevons Paradox: efficiency gains in using a resource can increase total demand for that resource, because the drop in cost expands how much the resource gets used.

The mechanism is ordinary economics: when the cost of doing something falls and demand for that thing is elastic (responsive to price), the quantity demanded rises by more than enough to offset the lower unit cost — so total consumption goes up.

Applied to AI and labor, the argument is: if AI makes producing a unit of work (a report, a feature, a design) cheaper, and if the market's appetite for that kind of work is elastic — more clients, more projects, more capability worth pursuing — then organizations that chase that expanded demand will need more people to direct and supervise the increased volume of AI-assisted output, not fewer. This does not happen automatically; it depends on demand actually being elastic and on organizations choosing to pursue growth with the savings rather than just cutting costs.

KEY POINTS

- → Jevons observed that more fuel-efficient steam engines led to more total coal consumption, not less.
- → The mechanism: lower cost per unit of use expands how many situations and users can afford to use it.
- → This only happens when demand is elastic — responsive enough to price that a drop in cost more than offsets the smaller unit cost.
- → Applied to AI: cheaper task execution can expand the total volume of work an organization takes on.
- → More volume, pursued deliberately, requires more people to direct and check that expanded output.

A simple elasticity model of the paradox

This is a general economic model, not a figure from the video, included to make the mechanism concrete: as price per unit falls, quantity demanded rises faster than price falls (because demand is elastic here), so total spend on the resource goes up even though each unit is cheaper — mirroring what Jevons observed with coal.

```
def demand(price, base_price=100, base_qty=100, elasticity=-1.5):
    """Constant-elasticity demand curve: Q = base_qty * (price / base_price) **
elasticity.
    elasticity < -1 means demand is 'elastic': a 1% price drop grows quantity by more
than 1%.
    """
    return base_qty * (price / base_price) ** elasticity

for price in [100, 75, 50, 25]:
    qty = demand(price)
    total_spend = price * qty
    print(f"price={price:>3}  quantity={qty:7.1f}  total_spend={total_spend:8.1f}")

# price=100  quantity=  100.0  total_spend= 10000.0
# price= 75  quantity=  153.9  total_spend= 11542.5
# price= 50  quantity=  282.8  total_spend= 14142.1
# price= 25  quantity=  800.0  total_spend= 20000.0
# As price per unit falls, both quantity used AND total spend rise -- the Jevons effect.
```

05 New role shapes: goal-setting and supervision

7:50

> As AI takes over how to execute a task, human value shifts toward deciding what should be done and why, then setting goals for AI and supervising its output — functions that used to be reserved for management.

Individual output used to be capped by how much a person could physically produce. A programmer's value was partly bounded by how many lines of correct code they could write in a day, which limited what a team could build. When AI can produce a first draft of that code far faster than a person could, the cap loosens, and the scarce, valuable activity shifts from producing the work to deciding what work is worth producing.

That shift moves 'what should we build and why' from a periodic planning exercise into an everyday part of more people's jobs. It also moves goal-setting and supervision — telling an AI system what outcome to aim for, then checking that it actually achieved that outcome correctly — down from being a purely managerial function into something individual contributors do routinely.

Hands-on work does not vanish; it becomes a smaller share of the job, with more time going to architecture-level thinking: what capability should exist, what will meet a need the customer has not yet articulated, and whether an AI's output actually satisfies that need.

KEY POINTS

→ Removing an execution bottleneck shifts value toward deciding what to build, not how to build it.

→ Goal-setting — defining the outcome an AI system should aim for — becomes a routine individual skill, not just a management function.

→ Supervision — checking that AI actually did what was intended, correctly — becomes equally routine.

→ Hands-on execution work shrinks as a share of the job without disappearing entirely.

→ The remaining scarce skill is judging whether an output meets a real, sometimes unstated, need.

A goal-and-constraints spec, instead of hand-written code

Instead of writing a rate limiter line by line, a developer now spends their effort defining the goal, the constraints the AI's implementation must satisfy, and how they will verify it — then reviews the AI's output against that spec rather than against their own memory of how to write it.

```
task: "Add rate limiting to the public API"
goal: "Prevent abuse without blocking legitimate bursty traffic"
constraints:
  - "No client exceeds 100 requests/min sustained"
  - "Allow short bursts up to 150 requests in any 10s window"
  - "Return 429 with a Retry-After header on violation"
acceptance_criteria:
  - "Unit tests pass for both burst and sustained-load scenarios"
  - "p99 latency overhead added by the limiter stays under 5ms"
review_focus:
  - "Does the AI's chosen algorithm actually satisfy the burst constraint?"
  - "Are edge cases handled: clock skew, multiple distributed nodes?"
```

06 The four skills the new roles require

9:57

> Four traits determine who moves up the diamond: flexibility, curiosity, creativity, and critical thinking — all judgment skills that AI does not replace.

Flexibility means staying adaptable as job definitions change under you. Nobody can fully predict where AI-driven change goes next: some existing jobs will persist largely unchanged, others will transform, and some brand-new roles will appear that don't exist yet. People who treat themselves as lifelong learners handle this shift better than people who expect to master one fixed skill set and stop.

Curiosity is the habit of asking what should we try next and why does this matter — the questions that lead to finding the next opportunity before it's obvious. Curious people tend to be the ones who spot where value is shifting early.

Creativity is what turns a curious question into something new: an idea, a design, a product direction. Curiosity finds the question; creativity produces an answer worth pursuing.

Critical thinking is treated as the most important of the four. It is the discipline of asking not just can we do this, but should we — whether a proposed action actually serves what stakeholders and clients need. As AI increasingly handles the 'how,' critical thinking becomes the main check on whether the 'what' and 'why' are actually correct, which is exactly the judgment AI cannot supply on its own.

KEY POINTS

- → Flexibility: adapting as job definitions shift, treating learning as continuous rather than front-loaded.
- → Curiosity: routinely asking what's next and why it matters, which surfaces opportunities early.
- → Creativity: converting a curious question into a concrete new idea or direction.
- → Critical thinking: asking whether something should be done, not only whether it can be — validating that outcomes actually meet real needs.
- → Of the four, critical thinking is framed as the most important because it is the check on everything AI produces.

A critical-thinking checklist before accepting AI output

Before shipping an AI-generated recommendation, a critical thinker asks: who is this actually for, what does it cost if this is wrong, does this match what was actually requested (not just what's plausible-sounding), and is there a simpler or more honest answer being skipped over?

07 Evidence, and the catch that decides who benefits

12:04

> IBM's stated plan to hire two to three times more entry-level people in 2026 despite heavy AI use is offered as evidence the growth pattern is real — but the benefit only shows up for organizations that choose growth over pure cost-cutting.

As supporting evidence, IBM is cited as planning to hire two to three times as many entry-level people in 2026 as in the prior year, while also using AI extensively. A company purely focused on cutting costs would do the opposite — shrink headcount and stop there. Hiring more entry-level people alongside heavy AI adoption is consistent with a company using AI-driven efficiency to pursue more growth, which is exactly the condition Jevons Paradox requires to produce more (not less) demand for people.

The catch is that this outcome is a choice, not a guarantee. The diamond effect only appears when an organization reinvests the savings from AI efficiency into more projects, more clients, or new capabilities. An organization that instead just banks the savings and keeps output flat does not generate the extra demand that would pull people up into better roles — it just gets the same output for a smaller team.

The practical implication for job descriptions: organizations that want to keep developing talent need to actively rewrite what entry-level roles look like to fit the new pattern, rather than assuming the market will sort it out. Individuals, in turn, need to actively build the four skills above rather than assume AI access will carry them up the diamond automatically.

KEY POINTS

- → IBM's stated plan to substantially increase entry-level hiring in 2026 while using AI heavily is used as a real-world data point for the growth pattern.
- → A cost-cutting-only strategy would predict shrinking headcount, the opposite of that hiring pattern.
- → The diamond/Jevons effect requires reinvesting efficiency savings into more output, not just banking them.
- → Companies that only cut costs do not generate the extra demand that pulls workers into higher-value roles.
- → Both organizations (job descriptions) and individuals (skills) have to act deliberately for the promotion to happen.

Two companies, same AI tools, different outcomes

Company A adopts AI purely to cut costs: same output, smaller team, savings banked. Demand for its people does not grow, and no one moves up the diamond. Company B adopts the same AI tools but reinvests the freed-up capacity into new markets and projects. Its people move into higher-judgment roles, and it needs more of them, not fewer, to direct the expanded scope.

Glossary

Jevons Paradox

The observation, first made in 1865 regarding coal and steam engines, that making a resource more efficient to use tends to increase total demand for it rather than reduce it, because falling costs expand how much it gets used.

Elasticity of demand

A measure of how much the quantity of something demanded changes when its price changes; when demand is elastic, a price drop increases quantity by more than enough to raise total spending.

Entry-level role

A job whose core function is to execute a specific, assigned task with limited independent decision-making authority.

Experienced-level role (individual contributor)

A job responsible for owning a complete, larger piece of work end-to-end, including how its parts fit together, rather than just executing one assigned fragment.

C-suite

The most senior executives in an organization (e.g., CEO, CTO, CFO), whose responsibility spans the entire organization rather than one function or team.

Upskilling

Deliberately developing new or higher-level skills, typically to take on more responsibility or adapt to a changed job requirement.

Pyramid vs. diamond org model

Two ways of picturing how many people occupy each level of responsibility in an organization: a pyramid is widest at the entry level and narrows steadily upward; a diamond is narrower at entry level but wider through the middle experience and senior bands.

AI supervision

The practice of setting a clear goal and constraints for an AI system, then checking its output against that goal to confirm it is correct and appropriate before it is used or shipped.

Check yourself

Answer first, then open each one.

Why does Jevons Paradox predict that AI could increase, rather than decrease, total demand for human labor, even though AI makes individual tasks cheaper to perform?

When the cost of doing a task falls and demand for that task is elastic, organizations don't just do the same amount of work for less money — they take on more of it (more clients, more projects). That expanded scope needs more people to direct and supervise it, so the paradox is that efficiency can raise total demand for the people who use the now-cheaper capability, rather than reduce the need for them.

What specifically distinguishes an experienced-level role from an entry-level role in this lesson's model, and why does AI make it easier for entry-level people to reach that level sooner?

Experienced-level roles own a larger, complete piece of work end-to-end (e.g., an entire subsystem and its integration) rather than just executing an assigned fragment. AI absorbing routine execution means one person directing AI tools can accomplish what used to require an army of entry-level people doing manual work, so a single person can credibly own that larger scope sooner, without first spending years doing the execution themselves.

Why won't a company that uses AI purely to cut costs see the same 'diamond' effect as a company that uses AI to grow?

The diamond/Jevons effect only appears when saved costs get reinvested into more output — more clients, more projects — which is what creates the extra demand for people to direct and supervise that output. A company that just banks the savings and shrinks its workforce keeps demand for labor flat or falling, so nothing pulls its people up into higher-value roles.

Why does this lesson treat critical thinking, rather than raw technical skill, as the most important skill in an AI-driven job market?

AI increasingly handles the 'how' of executing a task, so the differentiating human contribution becomes deciding 'whether and why' a task should be done at all, and validating that an AI's output actually serves stakeholders' real needs. That is a critical-thinking function, not an execution skill, and it is the one AI does not supply for itself.

Why can't an entry-level worker simply be dropped into a senior role just because AI has removed the routine tasks that used to occupy them?

Senior and experienced roles depend on judgment built from practicing decisions under real conditions -- knowing which risks matter, what 'good' output looks like, how to weigh trade-offs. AI can remove the busywork, but it cannot substitute for that calibration, which is why the lesson still points to internships and real experience, not just AI access, as the path upward.

Where to go next

→ Read William Stanley Jevons' 1865 argument in 'The Coal Question' for the original version of the efficiency-increases-demand claim.

- → Look up how ATM adoption affected the total number of US bank tellers from the 1970s through the 2000s, a well-documented non-AI example of the same paradox playing out (tellers per branch fell, but more branches opened and total teller employment grew for decades).
- → Once reported, compare IBM's actual 2026 entry-level hiring numbers against this lesson's claim to see whether the prediction held.
- → Practice writing a goal-and-constraints spec (like the rate-limiting example in this lesson) for a task you'd normally do by hand, then compare an AI's output against it and note what your review actually caught.
- → Pick one routine task in your own job that AI already does reasonably well, and write out concretely what supervising that task well would require of you.
